CURRICULUM VITAE

Roma Varghese, PhD

Earth system science · Ecohydrodynamics · Physical oceanography
romavarghese17@gmail.com

Research profile

Research on the environmental impacts of climate variability and change through monsoon rainfall, basin hydrology and terrestrial vegetation, with emphasis on India and monsoon Asia. Expertise includes tropical ocean teleconnections, satellite observations and AI applications in ecohydrology.

Research experience

Project Scientist-II · Indian Institute of Tropical Meteorology, Pune

Appointment began April 2024

Climate variability, prediction and fluid dynamics; decadal prediction of Indian river-basin hydrology.

POGO–SCOR visiting fellowship · JAMSTEC, Yokohama, Japan

September–November 2023

Sustained ocean observations and applications.

MTech dissertation · National Institute of Oceanography, Goa

July 2017–April 2018

Recent trends in chlorophyll-a in the southeastern Arabian Sea.

Education

PhD · Earth System Science and Technology

Indian Institute of Technology Kharagpur · July 2018–September 2024

Thesis: Tropical Ocean Teleconnections with Gross Primary Productivity of Monsoon-Asia with Emphasis over India.

MTech · Ocean Technology

Cochin University of Science and Technology · July 2016–April 2018

University first rank and gold medal · CGPA: 9.08

BTech · Civil Engineering

Cochin University of Science and Technology, Toc H Institute of Science and Technology · July 2012–April 2016 · CGPA: 8.27

Technical expertise

AI applications in ecohydrology; vegetation and hydrologic modelling; Python and R; QGIS, SNAP, ArcMap and ERDAS; Grapher and Origin. Additional training in architectural design, with experience in AutoCAD, Revit Architecture, 3ds Max and STAAD.Pro.

Fieldwork & professional development

Beach profiling along the Kerala coast; suspended sediment measurements at the mouth of the Mandovi Estuary, Goa; mangrove biophysical measurements in Bhitarkanika; and permanent 20 m × 20 m plots around a pheno-meteorological observation site in Similipal Biosphere Reserve, Odisha.

Science Journalism, Indian Science Communication Society, November 2024–November 2025 · Grade A+.

Publications

1. Varghese, R., Chowdary, J. S., & Gnanaseelan, C. (2026). Disentangling hydroclimatic and irrigation controls on vegetation photosynthesis variability in India. *Ecohydrology & Hydrobiology*. 10.1016/j.ecohyd.2026.100811
2. Varghese, R., Chowdary, J. S., & Gnanaseelan, C. (2026). Monsoon-driven hydrological pathways of flood generation in semi-arid river basins. *Hydrological Processes, 40(7)*. 10.1002/hyp.70622
3. Varghese, R., Behera, S., & Behera, M. D. (2025). An artificial intelligence-based framework for modeling vegetation photosynthesis anomalies in India corresponding to tropical oceanic variability. *Global Change Biology, 31(12)*. 10.1111/gcb.70636
4. Varghese, R., Behera, S., & Behera, M. D. (2024). Tropical ocean teleconnections with gross primary productivity of monsoon-Asia. *Science of The Total Environment, 938(5)*. 10.1016/j.scitotenv.2024.173337
5. Varghese, R., Behera, S. K., & Behera, M. D. (2023). Significant inverse influence of tropical Indian Ocean SST on SIF of Indian vegetation during the summer monsoon onset phase. *Remote Sensing, 15(7)*. 10.3390/rs15071756
6. Behera, M. D., Prakash, J., Paramanik, S., Mudi, S., Dash, J., Varghese, R., et al. (2022). Assessment of tropical cyclone Amphan affected inundation areas using Sentinel-1 satellite data. *Tropical Ecology, 63(1)*. 10.1007/s42965-021-00187-w
7. Varghese, R., & Behera, M. D. (2019). Annual and seasonal variations in gross primary productivity across the agro-climatic regions in India. *Environmental Monitoring and Assessment, 191, 631*. 10.1007/s10661-019-7796-2
8. Behera, S., Varghese, R., Ratnam, J. V., Doi, T., & Behera, M. D. (2025). Role of ocean and climate in terrestrial ecosystems. *Sustainable Development Perspectives in Earth Observation · Elsevier · pp. 419–435*. 10.1016/B978-0-443-14072-3.00002-2
9. Paramanik, S., Varghese, R., Behera, M. D., Barnwal, S., Behera, S. K., & Bhattyacharya, B. K. (2022). Photosynthetic Variables Estimation in a Mangrove Forest. *Advances in Remote Sensing for Forest Monitoring · John Wiley & Sons · pp. 126–149*. 10.1002/9781119788157.ch6

Selected conference contributions

- Indian Ocean–Monsoon Climate Workshop, IITM Pune (2026): artificial neural network modelling of hydroclimatic controls on vegetation photosynthesis.
- INTROMET, IITM Pune (2025): diagnostic analysis of the 2024 Krishna River basin flood.
- AGU Fall Meeting (2024): ANN-based predictability of vegetation photosynthesis anomalies associated with tropical Pacific SST variability.
- JpGU, EGU and the Biodiversity and Climate Change Workshop (2023): oceanic influences, hydroclimate and vegetation photosynthesis.
- AGU Fall Meeting (2022): oceanic influence on vegetation anomalies in India.

Awards & fellowships

- Best Poster Award, INTROMET, IITM Pune (2025).
- POGO–SCOR visiting fellowship (2023).
- AGU Fall Meeting Student Travel Grant (2022).
- IIT Kharagpur PhD institute fellowship (2018–2023).
- University first rank and gold medal, MTech in Ocean Technology, CUSAT.
- E-Grantz scholarship (2016–2018); Merit-cum-Means scholarship (2012–2013).